

samy kamkar - evercookie - virtually irrevocable persistent cookies

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Content

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evercookie

October 11, 2010: Reported on the front page of the [New York Times](#)

Find the latest details, code, and implementations on github @
<https://github.com/samyk/evercookie>

DESCRIPTION

evercookie is a javascript API available that produces extremely persistent cookies in a browser. Its goal is to identify a client even after they've removed standard cookies, Flash cookies (Local Shared Objects or LSOs), and others.

evercookie accomplishes [this](#) by storing the cookie data in several types of storage mechanisms that are available on the local browser. Additionally, **if** evercookie has found the user has removed any of the types of cookies in question, it recreates them [using](#) each mechanism available.

Specifically, when creating a **new** cookie, it uses the following storage mechanisms when available:**

- Standard [HTTP Cookies](https://en.wikipedia.org/wiki/HTTP_cookie)
- [HTTP Strict Transport Security (HSTS)](<https://sites.google.com/a/chromium.org/dev/Home/chromium-security/client-i>)
- [Local Shared Objects](https://en.wikipedia.org/wiki/Local_Shared_Object) (Flash Cookies)
- Silverlight [Isolated Storage](<https://www.silverlight.net/learn/quickstarts/isolatedstorage/>)
- Storing cookies in RGB values of auto-generated, force-cached PNGs [using](#) HTML5 Canvas tag to read pixels (cookies) back out
- Storing cookies in [Web History](<https://sa.my/csshack/>)
- Storing cookies in HTTP [ETags](https://en.wikipedia.org/wiki/HTTP_ETag)
- Storing cookies in [Web cache](https://en.wikipedia.org/wiki/Web_cache)
- [window.name](https://en.wikipedia.org/wiki/HTTP_cookie#window.name) caching
- Internet Explorer [userData]([https://msdn.microsoft.com/en-us/library/ms531424\(VS.85\).aspx](https://msdn.microsoft.com/en-us/library/ms531424(VS.85).aspx)) storage
- HTML5 [Session Storage](<https://dev.w3.org/html5/webstorage/#the-sessionstorage-attribute>)
- HTML5 [Local Storage](<https://dev.w3.org/html5/webstorage/#dom-localstorage>)
- HTML5 [Global Storage](<https://developer.mozilla.org/en/dom/storage#globalStorage>)
- HTML5 [Database Storage](<https://dev.w3.org/html5/webdatabase/>) via SQLite
- HTML5 [IndexedDB](<https://www.w3.org/TR/IndexedDB/>)
- Java [JNLP PersistenceService](<https://docs.oracle.com/javase/1.5.0/docs/guide/javaws/jnlp/index.html>)
- Java [CVE-2013-0422 exploit](<https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2013-0422>) (applet sandbox escapin

TODO:** adding support **for**:

- Caching in [HTTP Authentication](https://en.wikipedia.org/wiki/Basic_access_authentication)
- **Using** Java to produce a unique key based off of NIC info
- Google Gears

Got a crazy idea to improve [this](#)? [Email me!](mailto:code@sa.my)

EXAMPLE

****Cookie found:** *uid* = currently not set**

Click to create an evercookie. Don't worry, the cookie is a random number between 1 and 1000, not enough for me to track you, just enough to test evercookies.

Now, try deleting this "uid" cookie anywhere possible, then

or

DOWNLOAD

evercookie is written in JavaScript and contains portions in Java, SWF/ActionScript (Flash) and C# (Silverlight).

Some backend pieces in PHP, but also available in [Node.js](https://github.com/truongsinh/node-evercookie) and [Django](https://github.com/truongsinh/django-evercookie).

Get the latest source from github: https://github.com/samyk/evercookie

FAQ

****What is the point of evercookie?****

Evercookie is designed to make persistent data just that, persistent. By storing the same data in several locations that a client can access, **if** any of the data is ever lost (**for** example, by clearing cookies), the data can be recovered and then reset and reused.

Simply think of it as cookies that just won't **go** away.

****PRIVACY CONCERN! How **do** I stop websites **from** doing **this**?**

Great question. So far, I've found that **using** [Private Browsing](https://www.apple.com/pro/tips/privacy_safari.html) in [Safari](<https://www.apple.com/safari/>) will stop ALL evercookie methods after a browser restart.

****What **if** the user deletes their cookies?**

That's the great thing about evercookie. With all the methods available, currently thirteen, it only takes one cookie to remain **for** most, **if** not all, of them to be reset again.

For example, **if** the user deletes their standard HTTP cookies, LSO data, and all HTML5 storage, the PNG cookie and history cookies will still exist. Once either of those are discovered, all of the others will come back (assuming the browser supports them).

****Why not **use** EFF's [Panopticlick](<https://panopticlick.eff.org/>)?**

Panopticlick is an awesome idea, however the uniqueness really only helps in consumer machines and typically not systems running in a business or corporation. Typically those systems are virtually identical and provide no difference in information where a home user's laptop would. Evercookie is meant to be able to store the same unique data a normal cookie would.

****Does **this** work cross-browser?**

If a user gets cookie'd on one browser and switches to another browser, as long as they still have the Flash Local Shared Object cookie, the Silverlight Isolated Storage, the Java JNLP PersistenceService or the Java CVE-2013-0422 exploit cookie, the cookie should reproduce in both browsers.

****Does the client have to install anything?**

No, the client simply uses the website without even knowing about the persistent data being set, just as they would **use** a website with standard HTTP cookies.

****Does the server have to install anything?**

The server must at least have access to the JavaScript evercookie file. Additionally, to [use](#) Local Shared Object (Flash Cookies) storage, the evercookie.swf file must be present, and to [use](#) the auto-generated PNG caching, standard caching and ETag storage mechanisms, PHP must be installed and evercookie_(png|etag|cache).php must be on the server.

All of these are available in the download.

****Is evercookie open source?****

Yes, evercookie is open source. The code is in readable format without any obfuscation. Additionally, the PHP files are open source as is the FLA (Flash) code used to generate the SWF Flash object. You can compile the Flash object yourself or [use](#) the pre-compiled version (evercookie.swf).

****How does the PNG caching work?****

When evercookie sets a cookie, it accesses evercookie_png.php with a special HTTP cookie, different than the one used [for](#) standard session data. [This](#) special cookie is read by the PHP file, and [if](#) found, generates a PNG file where all the RGB values are set to the equivalent of the session data to be stored. Additionally, the PNG is sent back to the client browser with the request to cache the file [for 20 years](#).

When evercookie retrieves [this](#) data, it deletes the special HTTP cookie, then makes the same request to the same file without any user information. When the PHP script sees it has no information to generate a PNG with, it returns a forged HTTP response of "304 Not Modified" which forces the web browser to access its local cache. The browser then produces the cached image and then applies it to an HTML5 Canvas tag. Once applied, evercookie reads each pixel of the Canvas tag, extracting the RGB values, and thus producing the initial cookie data that was stored.

****How does the Web History storage work****

When evercookie sets a cookie, assuming the Web History caching is enabled, it Base64 encodes the data to be stored. [Let's](#) assume [this](#) data is "bcde" in Base64. Evercookie then accesses the following URLs in the background:

```
google.com/evercookie/cache/b
google.com/evercookie/cache/bc
google.com/evercookie/cache/bcd
google.com/evercookie/cache/bcde
google.com/evercookie/cache/bcde-
```

These URLs are now stored in history.

When checking [for](#) a cookie, evercookie loops through all the possible Base64 characters on google.com/evercookie/cache/, starting with "a" and moving up,

but only **for** a single character. Once it sees a URL that was accessed, it attempts to brute force the next letter. **This** is actually extremely fast because ****no requests**** are made to the server. The history lookups are simply locally in JavaScript **using** the [CSS History Knockout](https://sa.my/csshack/). *Evercookie knows it has reached the end of the **string** as soon as it finds a URL that ends in **"-"**.*

USAGE

```
**<script type="text/javascript" src="evercookie.js"></script>

<script>
var ec = new evercookie();**

// set a cookie "id" to "12345"
// usage: ec.set(key, value)
**ec.set("id", "12345");**

// retrieve a cookie called "id" (simply)
**ec.get("id", function(value) { alert("Cookie value is " + value) });**

// or use a more advanced callback function for getting our cookie
// the cookie value is the first param
// an object containing the different storage methods
// and returned cookie values is the second parameter
**function getCookie(best_candidate, all_candidates)
{
    alert("The retrieved cookie is: " + best_candidate + "\n" +
        "You can see what each storage mechanism returned " +
        "by looping through the all_candidates object.");

    for (var item in all_candidates)
        document.write("Storage mechanism " + item +
            " returned: " + all_candidates[item] + "<br>");
}
ec.get("id", getCookie);**

// we look for "candidates" based off the number of "cookies" that
// come back matching since it's possible for mismatching cookies.
// the best candidate is most likely the correct one
**</script>**
```

SEE ALSO

[csshack](https://sa.my/csshack/), [best website ever](https://sa.my)

BUGS

See **CONTACT**.

CONTACT

Questions or comments, email me: **code@sa.my**.

Visit sa.my *for more awesome stuff.*

evercookie, by **samy kamkar**, 2010/09/20

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